Task 7. A Physics-Informed Preprocessing Approach for Dark Matter Substructure Classification in Gravitational Lensing

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Abstract. We present a physics-informed preprocessing method for the classification of dark matter substructure in strong gravitational lensing images. Our approach exploits a fundamental property of the Singular Isothermal Sphere (SIS) lensing model: under an idealized smooth potential, the surface brightness of an Einstein ring is radially symmetric by construction. Any deviation from this symmetry must therefore originate from perturbations to the lensing potential induced by dark matter substructure. We formalize this observation through the Radial Symmetry Residual (RSR), a deterministic, parameter-free map that computes the per-pixel residual between the observed image and its radially averaged counterpart. The RSR channel is concatenated with the raw image to form a two-channel input to a standard ResNet-18 backbone, with no modifications to the training objective, optimizer, or hyperparameters relative to a single-channel baseline. Evaluated on a balanced dataset of 37,500 synthetic gravitational lensing images spanning three classes - No Substructure, SubHalo (CDM), and Vortex (axion) - the proposed model achieves a validation accuracy of 93.65%, a macro F1 score of 0.9357, and a macro-average AUC of 0.9915 under the one-vs-rest protocol. Grad-CAM visualizations confirm that the model attends to physically meaningful regions: diffuse arc activations for No Substructure, compact flux anomaly spots for SubHalo, and extended non-radial patterns for Vortex - in each case consistent with the corresponding RSR signal. These results demonstrate that injecting a classical physics prior at the input level, without any learnable physics module or auxiliary loss, constitutes an effective and interpretable strategy for dark matter substructure classification.

1 Dataset

The dataset comprises 37,500 single-channel gravitational lensing images of spatial resolution 150×150 pixels, partitioned into a training split of 30,000 samples and a validation split of 7,500 samples, following an 80/20 ratio. Each image is assigned to one of three mutually exclusive classes: No Substructure, SubHalo (Sphere), and Vortex. The class distribution is perfectly balanced across both splits, with each category contributing equally to the total sample count (10,000 training and 2,500 validation samples per class), thereby eliminating class imbalance as a confounding factor in model evaluation. All images were generated synthetically using the `lenstronomy` Python package, a well-established gravitational lensing simulation framework. A summary of the dataset composition is provided in Table 1. More information about the dataset, see EDA Notebook.

Split	Total Samples	% of Dataset	Class: No	Class: Sphere	Class: Vort
Train	30,000	80 %	10,000	10,000	10,000
Validation	7,500	20 %	2,500	2,500	2,500

Table 1: Dataset summary.

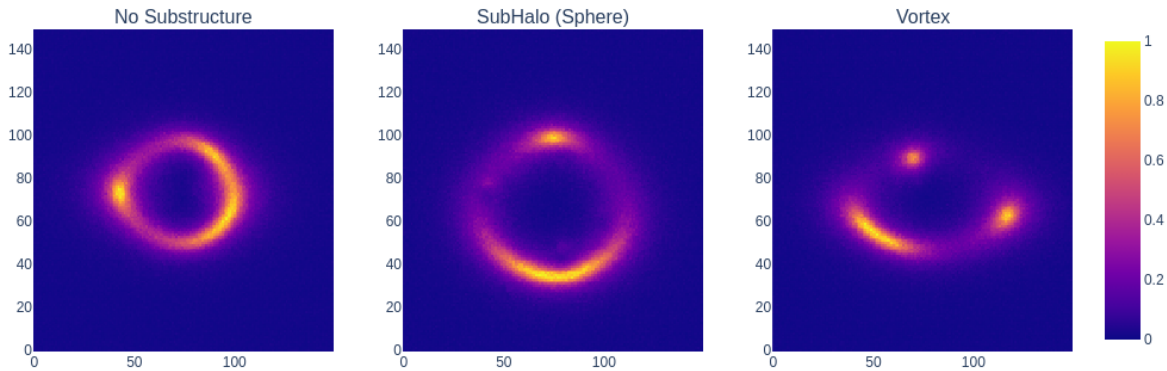

Figure 1: Example of a No Substructure, a Sphere, and a Vortex image.

2 Model

2.1 Radial Symmetry Residual

A gravitational lens well described by a Singular Isothermal Sphere (SIS) profile produces a deflection field that is, by construction, purely radial. Under this idealized model, the surface brightness of the lensed arc depends only on the angular distance from the optical axis, making the observed Einstein ring radially symmetric to first order. Any departure from this symmetry must therefore originate from perturbations to the smooth lensing potential - precisely the signature of dark matter substructure.

This observation motivates a deterministic, parameter-free preprocessing map that we call the Radial Symmetry Residual (RSR). Given a normalized single-channel image $I \in [0,1]^{H \times W}$, we first compute its radially averaged counterpart $\hat{I}_{\text{radial}}$ by partitioning the image plane into $N_{\text{bins}} = 75$ concentric annuli centered at the image centroid (c_x, c_y). For each pixel (x, y), its radial bin index is

$$k(x,y) = \left\lfloor \frac{r(x,y)}{r_{\max}} \right\rfloor$$

where

$$r(x,y) = \sqrt{(x-c_x)^2 + (y-c_y)^2}$$

and the mean intensity within each annulus k is

$$\mu_k = \frac{1}{|A_k|} \sum_{(x,y) \in A_k} I(x,y)$$

The radially averaged image $\bar{I}_{\text{radial}}$ is then reconstructed by assigning $\mu_{k(x,y)}$ to every pixel, yielding the best SIS-consistent approximation of I without any learned parameters. The RSR map is defined as the residual

$$S = \mathcal{N}(I - \bar{I}_{\text{radial}})$$

where $\mathcal{N}$ denotes min-max normalization to $[0, 1]$ applied per image. This operation is fully differentiable and admits an efficient GPU implementation via `scatter_add`, with a computational overhead negligible relative to the forward pass of ResNet-18.

The physical interpretation of S is class-dependent and directly exploitable for classification. For No Substructure images, the lensing potential is smooth and the arc closely follows the SIS prediction, so $S \approx 0$ almost everywhere. For SubHalo (Sphere/CDM) images, massive compact subhalos locally perturb the deflection angle, producing flux anomalies at discrete positions along the arc; S consequently exhibits localized high-intensity spots at those positions. For Vortex (axion) images, the quantum pressure of ultra-light axion dark matter generates extended topological defects - vortices in the condensate wavefunction - that imprint non-radial, spatially extended patterns onto the arc; S reflects this as diffuse, structured residuals distributed across the image. These qualitative differences are visible in Figure 2, which displays representative examples of S for each class.

Rather than feeding S as a standalone input, we concatenate it channel-wise with the original image, forming a two-channel tensor $[I, S] \in \mathbb{R}^{2\times H\times W}$. This design allows the backbone to jointly exploit raw photometric information and the physics-derived symmetry-breaking signal, while leaving the classification objective and all training hyperparameters identical to the common test baseline. No auxiliary loss terms, no learnable physics modules, and no additional hyperparameter tuning are required - the physical prior is injected entirely through the input representation.

2.2 Architecture used

The classification backbone is a ResNet-18 architecture, identical to the one employed in the common test baseline. The sole structural modification concerns the first convolutional layer, which is adapted to accept the two-channel input tensor $[I, S] \in \mathbb{R}^{2\times H\times W}$ described in Section 2.1. Specifically, `conv1` is redefined with `in_channels=2`, with weights initialized via Kaiming normal initialization. All remaining layers, including the four residual stages and the final fully connected head mapping to three output logits, are kept unchanged. The total parameter count is therefore virtually identical to the baseline.

3 Training

The model is optimized using AdamW with a learning rate of 1.020×10^{-4} and a weight decay of 1.370×10^{-4}, both determined via Optuna TPE search on the common test baseline and reused without modification - a deliberate choice to isolate the effect of the RSR input channel from any confounding hyperparameter tuning. No learning rate scheduler is applied. Training runs for 100 epochs with a batch size of 64 for training and 128 for validation, and the best checkpoint is selected according to minimum validation loss, achieved at epoch 97.

Split	**Loss**	**Accuracy**	**Macro F1**
Train	0.1552	0.9420 %	0.9418
Validation	0.1701	0.9365 %	0.9357

Table 2: Performance at best checkpoint (epoch 97).

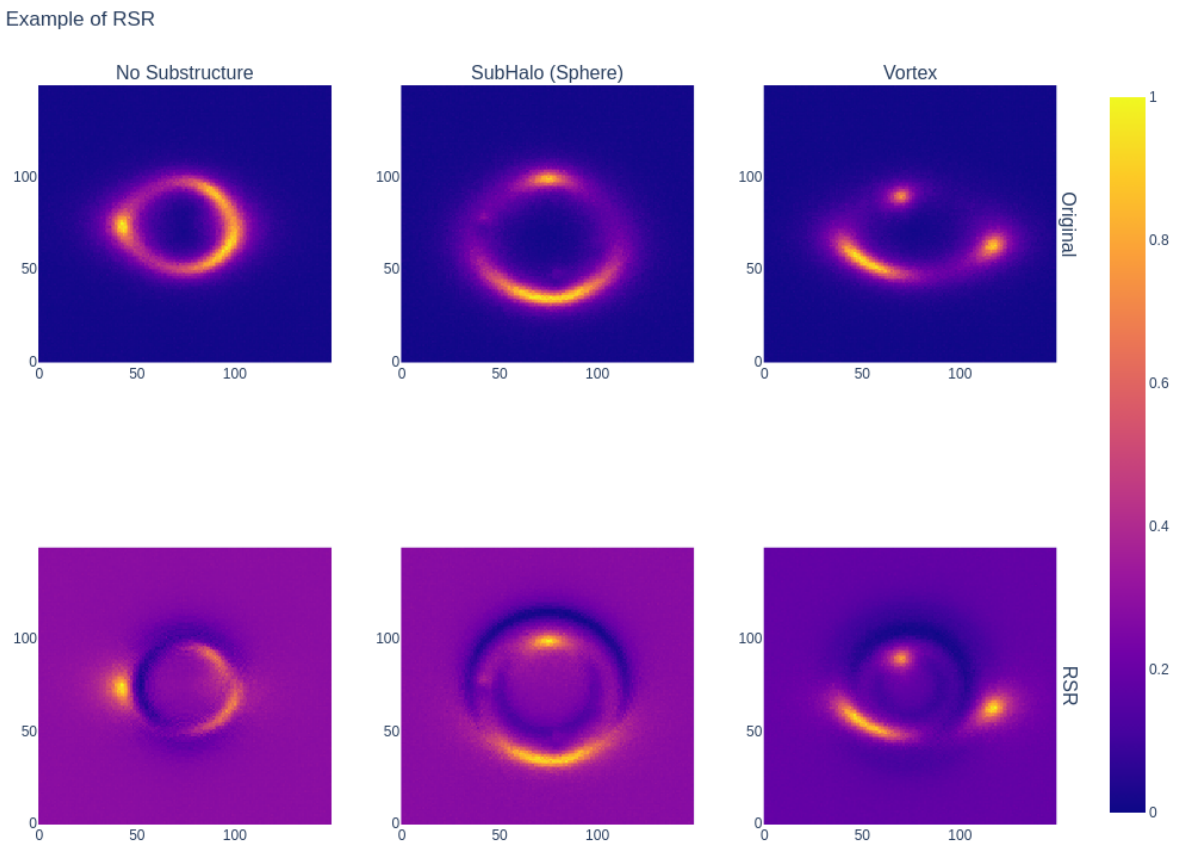

Figure 2: Example of a No Substructure, a Sphere, and a Vortex image using Radial Symmetry Residual.

The small gap between training and validation metrics indicates that the model generalizes well without overfitting, consistent with the learning curves shown in Figure 3. The absence of a scheduler did not hinder convergence - validation accuracy stabilizes above 0.93 by approximately epoch 60 and remains there for the remainder of training, suggesting that the fixed learning rate is well-suited to the loss landscape induced by the two-channel input.

4 Results

The model achieves strong discriminative performance across all three classes, as summarized by the confusion matrix in Figure 4 and the ROC curves in Figure 5. The confusion matrix reveals a clear asymmetry in classification difficulty. No Substructure is identified with near-perfect precision, with 99% of validation samples correctly classified and essentially zero confusion with the remaining classes. Vortex follows closely with a 96% recall, and the small fraction of misclassified samples are distributed almost exclusively toward No Substructure. SubHalo is the most challenging class, achieving 85% recall, with the primary source of error being confusion with No Substructure (8%) and, to a lesser extent, Vortex (6%). This pattern is physically interpretable: SubHalo perturbations are compact and localized, and for low-mass subhalos the deviation from a smooth SIS arc can be

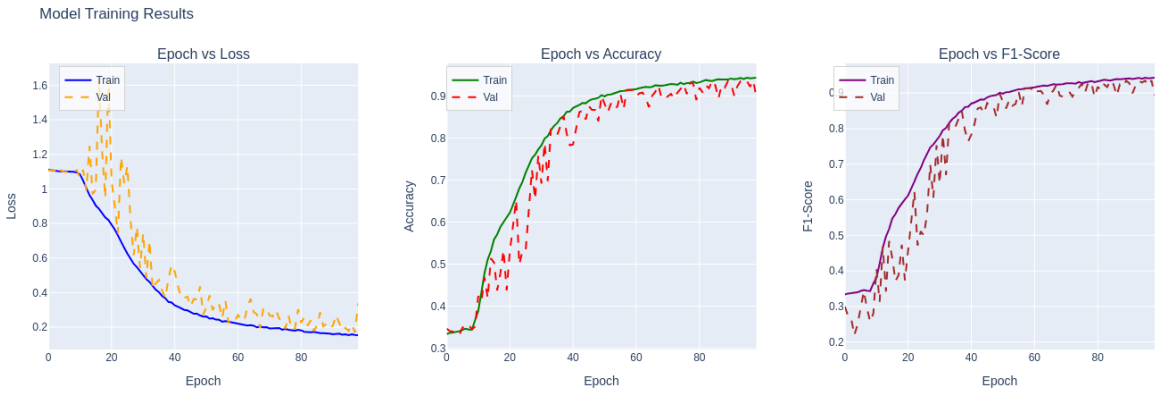

Figure 3: Epoch vs Loss, Accuracy, and Macro F1 during the training.

subtle enough that both the raw image and the RSR channel S produce a signal close to the No Substructure regime.

The ROC curves corroborate this picture quantitatively. All three classes achieve AUC scores above 0.985, with No Substructure at 0.9941, Vortex at 0.9951, and SubHalo at 0.9854, yielding a macro-average AUC of 0.9915. The curves rise steeply toward the top-left corner of the ROC space, confirming that the classifier maintains high true positive rates at very low false positive rates for all classes - a particularly relevant operating regime for a scientific application where false detections of dark matter substructure carry physical consequences. The lower AUC for SubHalo relative to the other two classes is consistent with the confusion matrix finding and reflects the inherent difficulty of detecting CDM subhalos from single lensing images: their signal is spatially localized and can mimic the absence of substructure when the subhalo mass is near the detection threshold of the RSR map.

5 Grad-CAM

To probe what spatial regions drive the model's decisions, we apply Gradient-weighted Class Activation Mapping (Grad-CAM) to the final convolutional layer of the ResNet-18 backbone (`layer4[-1].conv2`). For each validation sample, the Grad-CAM map is computed with respect to the ground-truth class index, producing a spatial heatmap that highlights the image regions whose activations contribute most positively to the predicted class score. The resulting overlays are shown in Figure 6, with one representative sample per class. The qualitative patterns are strikingly consistent with the physical interpretation of the RSR channel. For No Substructure, the activation is distributed broadly and uniformly along the Einstein ring, with no preferred location. This is the expected behavior: in the absence of substructure, all segments of the arc carry equivalent discriminative information, and the model correctly treats the ring as a whole rather than fixating on any particular region. The RSR channel for this class is correspondingly diffuse, showing only the faint residual noise of a nearly perfectly symmetric arc.

For SubHalo (Sphere), the Grad-CAM overlay concentrates activation at two spatially compact, high-intensity spots that coincide precisely with the positions of anomalous flux in the RSR channel. These spots correspond to the locations where CDM subhalos have locally perturbed the deflection angle, producing brightness excesses that deviate from the radially symmetric SIS prediction. The

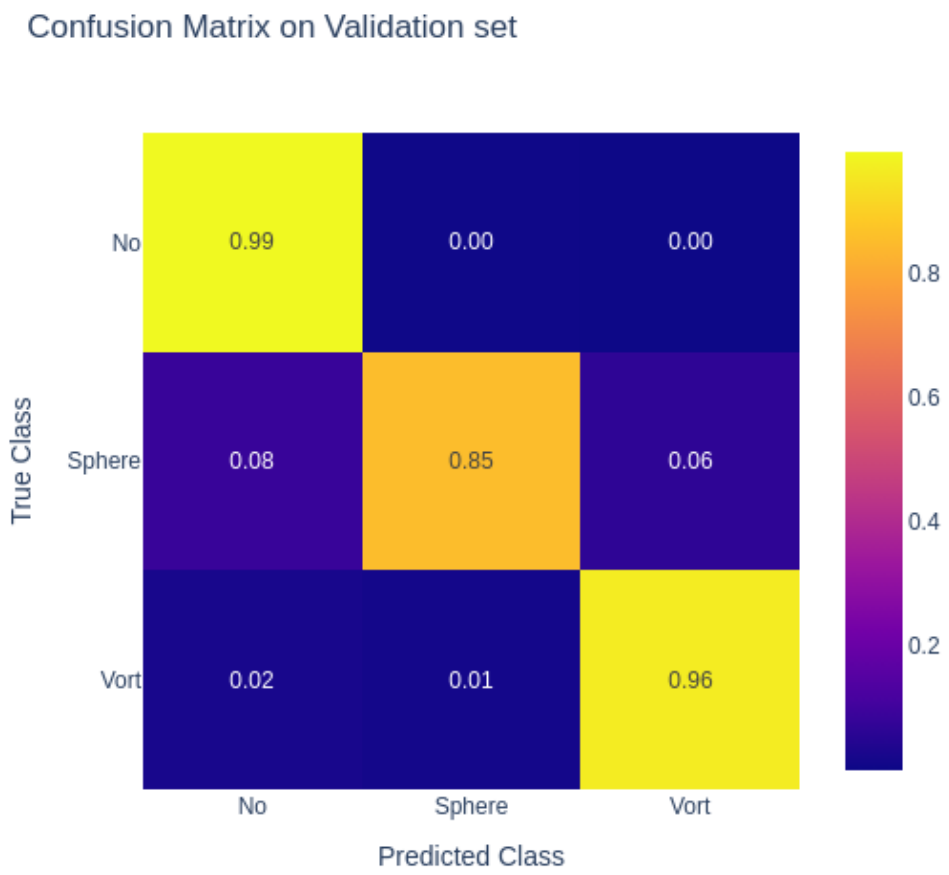

Figure 4: Confusion Matrix on Validation set.

model has learned to attend to exactly those positions - not to the overall arc morphology - which demonstrates that the RSR preprocessing successfully surfaces the physically relevant signal and guides the classifier toward it. For Vortex, the activation pattern is qualitatively distinct from both other classes. Rather than a single contiguous arc or isolated spots, the Grad-CAM map highlights two spatially separated, elongated regions distributed along the arc at positions where the RSR channel exhibits its characteristic non-radial modulation. This extended, multi-focal attention pattern is consistent with the topological nature of axion vortex perturbations, which imprint phase-coherent structures across a larger angular extent than CDM subhalos. Taken together, the Grad-CAM analysis provides interpretable evidence that the model does not rely on spurious correlates or global image statistics, but instead responds to physically meaningful local deviations from radial symmetry - precisely the information encoded in the RSR channel S. This alignment between model attention and physical signal constitutes the primary explainability contribution of this work, and supports the use of RSR as a principled, interpretable preprocessing stage for gravitational lensing classification.

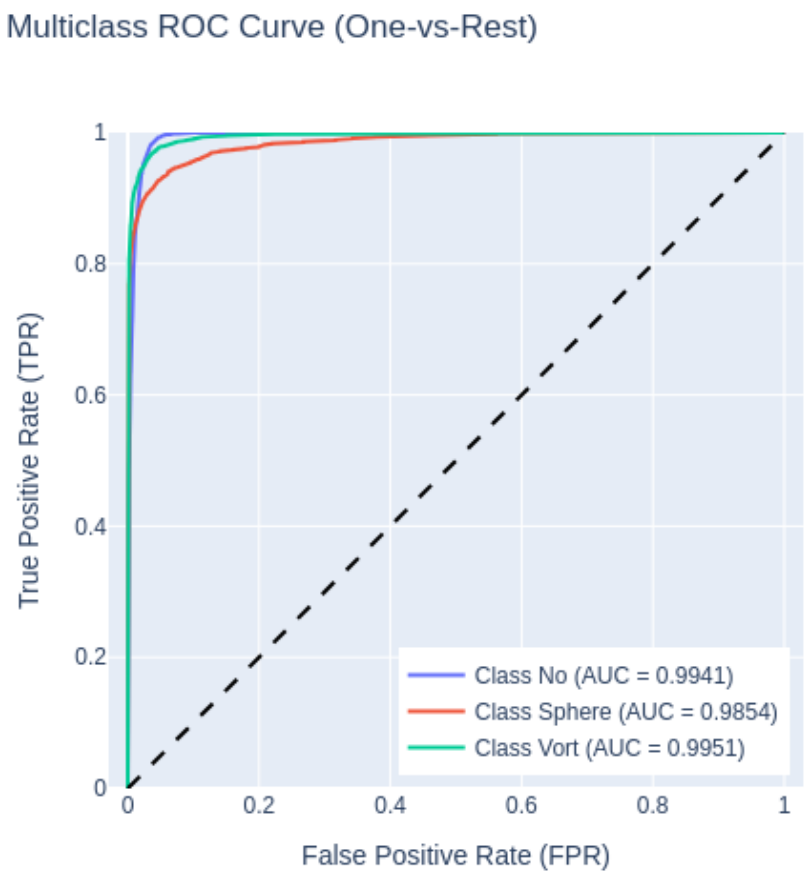

Figure 5: Multiclass ROC curve under the One-vs-Rest (OvR).

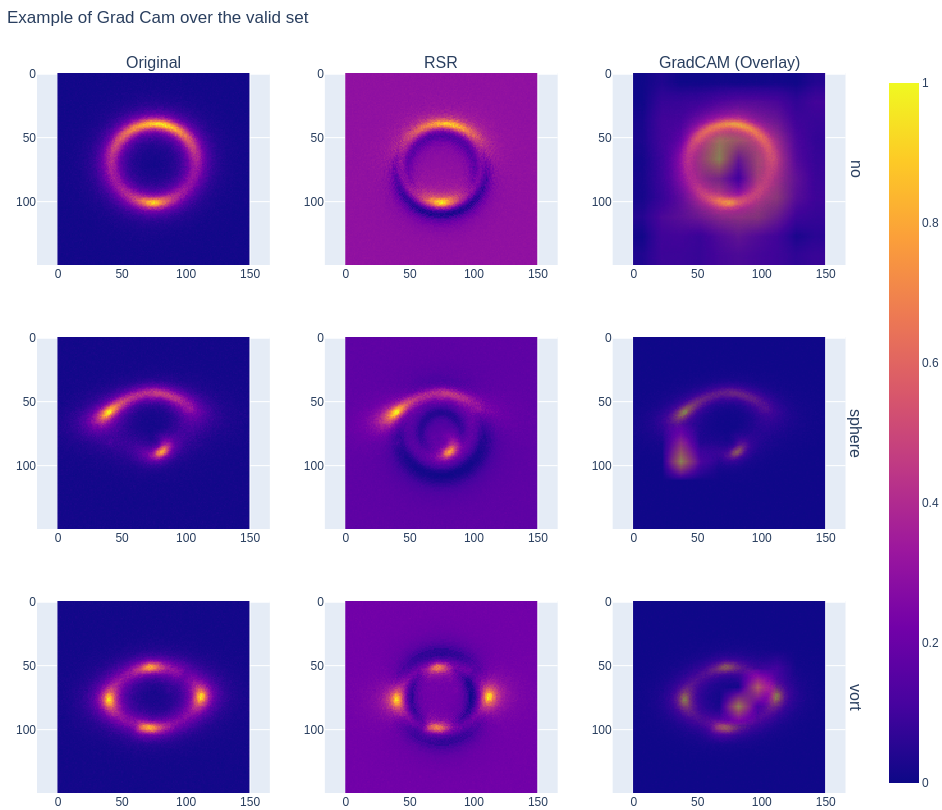

Figure 6: Example of 3 samples with Grad-Cam.